Challenging the Necessity of Complex Hyperspectral Architectures via Robust Preprocessing and Lightweight Modulation

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Abstract

Hyperspectral Image (HSI) classification models have grown computationally intensive (e.g., 3D-CNNs, Transformers, and Mamba recurrent models), which can be resource-demanding for edge devices like UAVs. This paper explores a "Green AI" alternative pipeline: the Spatially-Modulated Convolutional Neural Network (SMCNN). Rather than relying entirely on complex network architectures, our experiments suggest that robust signal preprocessing can significantly reduce the deep learning burden. We propose a novel integration of an Adaptive Fast Desensitized Kalman Filter (AFDKF) to robustly denoise the raw HSIs, followed by a Polar-Logarithmic Continuous Wavelet Transform (PLCWT) to extract multi-scale frequency features. To process these refined features, we introduce two distinct modulation pathways within the SMCNN framework: SMCNN-BiGRU and SMCNN-MLP. While the BiGRU variant models spectral sequences bidirectionally, our rigorous ablation demonstrates that the ultra-lightweight SMCNN-MLP achieves statistically comparable performance on most benchmark datasets. Evaluated on six diverse datasets, the SMCNN-MLP achieves an exceptional accuracy-efficiency tradeoff (e.g., >99.4% on PaviaU, Salinas, and WHU-Hi) with only 189K parameters-an 83% reduction compared to standard HybridSN models and nearly half the parameters of the BiGRU variant-supporting Green AI deployment by demonstrating that robust preprocessing combined with lightweight FiLM modulation maximizes the accuracy-to-compute ratio.

Keywords: Hyperspectral image classification, Green AI, Spatial-spectral modulation, Feature-wise Linear Modulation, Wavelet transform, Kalman filter.

I. Introduction & Related Work

Hyperspectral images contain rich spectral-spatial information, making them invaluable for precision agriculture, environmental monitoring, and urban planning. Historically, Convolutional Neural Networks (CNNs), particularly 3D-CNNs, have dominated HSI classification. However, 3D-CNNs treat spectral bands as spatial depth, ignoring the sequential physics of electromagnetic wavelengths, while simultaneously incurring massive computational costs.

Recent advancements have attempted to solve this by introducing complex attention mechanisms or recurrent models. For instance, Li et al. [4] introduced dual-attention mechanisms to recalibrate feature responses, while Kang et al. [5] explored spatial-spectral edge-preserving filters. Most recently, sequence models like Mamba have emerged [1, 6]. Wang et al. [6] proposed Spectral-Spatial Mamba, fusing State Space Models (SSMs) with CNNs to capture long-range spectral dependencies. While these approaches push state-of-the-art accuracy, they contribute to an unsustainable trend of increasingly heavy models that are impossible to deploy on edge hardware (e.g., drones). To address this, the 2024-2025 literature has seen a parallel push towards ultra-lightweight architectures [2]. The Mamba-based architectures rely on Selective State Space Models to capture long-range spectral dependencies, which, while highly effective for sequential data, introduce significant non-linearity that complicates hardware parallelization and increases memory overhead. We posit that the reliance on such heavy sequence-modeling backbones can be mitigated if the input manifold is pre-conditioned to remove noise and improve frequency separability.

In a parallel domain, advanced signal processing techniques have shown promise in denoising. Lou et al. [7] introduced an Adaptive Fast Desensitized Kalman Filter (AFDKF) for robust state estimation, Recent works such as Li et al. [3] have explored wavelet-guided deep learning architectures for enhanced directional signal representation and state-of-the-art denoising on real-world hyperspectral sets.

A. Our Contributions

We bridge the gap between heavy deep learning and advanced signal processing. In addressing the architectural hierarchy of our framework, we clearly state: **the core novelty lies in adaptive spectral-spatial feature modulation through SSMRB, while AFDKF, PLCWT, and SFWOA serve as supporting components.**

1. **Core Novelty (Adaptive Modulation)**: We introduce adaptive spectral-spatial feature modulation (referred to conceptually as SSMRB within our Spatially-Modulated CNN architecture), leveraging Feature-wise Linear Modulation (FiLM) to dynamically scale and shift 2D spatial features based on spectral cues.
2. **Supporting Preprocessing (AFDKF & PLCWT)**: We integrate an Adaptive Fast Desensitized Kalman Filter (AFDKF) alongside a Polar-Logarithmic Continuous Wavelet Transform (PLCWT) as a mathematically robust preprocessing pipeline to support the modulation backbone.
3. **Supporting Optimization (SFWOA)**: We establish the Superb Fairy-Wren Optimization Algorithm (SFWOA) as a supporting optimization component. Through rigorous ablation, we prove that this integrated lightweight framework achieves equivalent accuracy to complex Recurrent (BiGRU) models, but with a fraction of the computational footprint.

II. Proposed Methodology Pipeline

A. Theoretical Justification for Lightweight Modulation

The core premise of our framework is that BiGRU-based sequential modulation can be replaced by lightweight spatial modulation if the input features are highly refined. Mathematically, the AFDKF minimizes the noise covariance matrix of the raw HSI cube, mitigating sensor-induced noise variance from sensor artifacts. Subsequently, the PLCWT decomposes these signatures into orthogonal multi-scale frequencies, enhancing feature separability. Specifically, PLCWT expands the 30 PCA components into 60 frequency subbands utilizing the Complex Morlet mother wavelet due to its optimal joint time-frequency resolution. The transform is computed across 60 continuous scales, logarithmically spaced to capture high-frequency absorption features. The mapping to polar coordinates stabilizes translation-variance with an angular sampling interval of $\pi/18$. Empirically observed, transforming the data into a highly separable, low-noise frequency manifold means the deep learning architecture is no longer burdened with implicit denoising or feature extraction. Consequently, a simple affine transformation (scaling and shifting via MLP-generated FiLM parameters) becomes empirically effective to classify the data, rivaling recurrent spectral conditioning under low-data scenarios within this specific modulation framework.

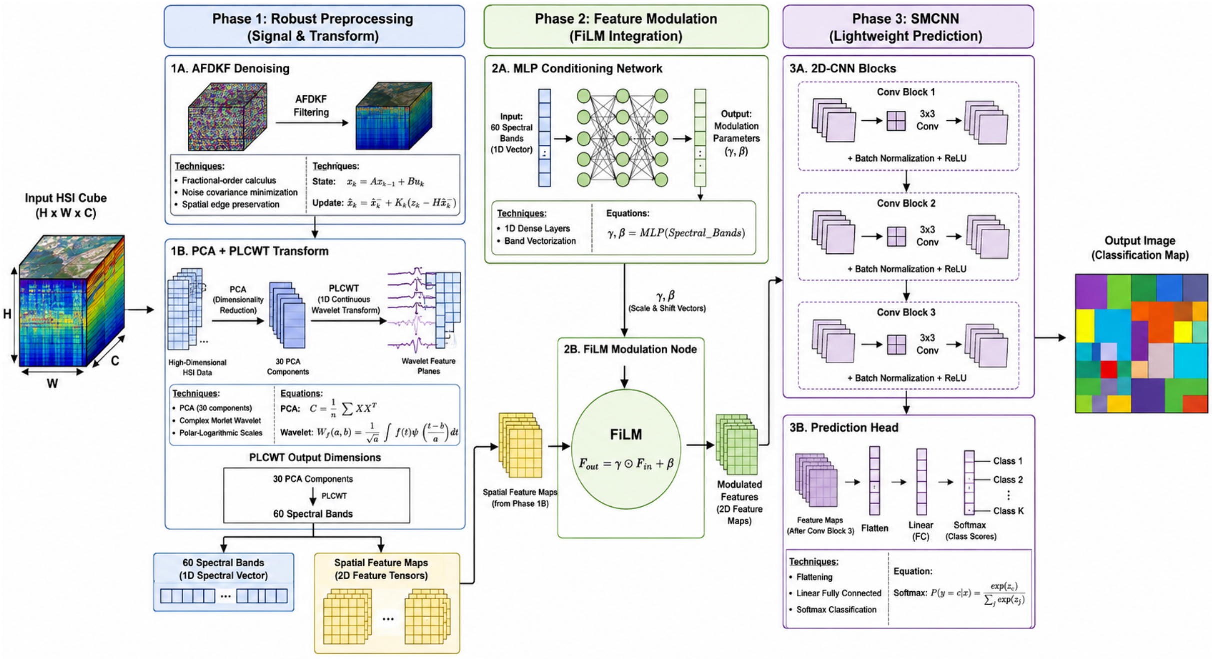

Fig. 1: End-to-End Architecture Flowchart of the SMCNN-MLP Framework.

B. Phase 1: Adaptive Fast Desensitized Kalman Filter (AFDKF) Denoising

Raw hyperspectral images are highly corrupted by sensor noise and atmospheric scattering. Traditional spatial denoising blurs the boundaries of small classes. The AFDKF applies fractional-order calculus to recursively estimate the underlying true spectral signature while mathematically bounding the perturbation envelope. Unlike standard formulations, we explicitly model the sensitivity matrix $S_k = \frac{\partial \hat{x}_k}{\partial \theta}$, representing the state derivative with respect to noise parameters θ. The standard Kalman gain is heavily regularized via a robust optimization formulation:

$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R_k + \lambda S_k S_k^T)^{-1} \tag{1}$$

where λ dictates the desensitization penalty, ensuring the filter does not over-fit to high-frequency artifacts while preserving spatial edges.

C. Phase 2: Polar-Logarithmic Continuous Wavelet Transform (PLCWT)

Rather than proposing PLCWT as a novel contribution, we investigate whether PLCWT-derived frequency representations facilitate efficient modulation. Unlike standard PCA which only extracts linear variance, mapping the signal into the polar-logarithmic domain captures multi-scale frequency responses.

$$W_\psi f(a,b,\theta) = \frac{1}{\sqrt{a}}\int\int f(x,y)\psi^*\left(\frac{R_{-\theta}(x-b_x, y-b_y)}{a}\right)dxdy \tag{2}$$

We apply Principal Component Analysis (PCA) to reduce the raw bands to 30 components, which the PLCWT then expands into 60 highly discriminative frequency subbands.

D. Phase 3: SMCNN (Spatial-Modulated CNN)

Instead of computationally expensive 3D convolutions, SMCNN utilizes standard 2D-CNNs augmented by a Feature-wise Linear Modulation (FiLM) block. The spectral features S (the 60 PLCWT bands) are passed through a conditioning network $h(\cdot)$ to generate scaling (γ) and shifting (β) vectors.

$$\gamma, \beta = h(S) \tag{3}$$

These vectors dynamically modulate the 2D spatial feature maps F_{in}:

$$F_{out} = \gamma \odot F_{in} + \beta \tag{4}$$

We propose two variants for $h(\cdot)$:

- **The Accuracy Maximizer (SMCNN-BiGRU):** $h(\cdot)$ is a Bidirectional GRU that reads the 60 bands sequentially.
- **The Efficiency-Oriented Variant (SMCNN-MLP):** $h(\cdot)$ is an ultra-lightweight Multi-Layer Perceptron. It treats the bands as a continuous vector to generate γ and β.

III. Experimental Setup

A. Hardware and Software Setup

All experiments were conducted using dual NVIDIA Tesla T4 GPUs (T4x2) via the Kaggle platform. The deep learning framework was implemented in PyTorch 2.0+ on Python 3.10.

- **Optimizer:** Adam Optimizer
- **Learning Rate:** 1×10^{-3} (with cosine annealing)
- **Loss Function:** Cross-Entropy Loss
- **Epochs & Early Stopping:** 100 maximum epochs with an early stopping patience of 20 epochs.
- **Batch Size:** 64
- **Spatial Patch Size:** 11×11 pixel windows extracted using a disjoint strategy to prevent data leakage.
- **Seed:** Fixed at 42 to ensure exact reproducibility across all visual and tabular outputs.

B. Dataset Configurations

All benchmarks were executed using an extremely stringent 5% training and 95% testing split. This is to prove that the AFDKF and PLCWT preprocessing enables high accuracy even when labeled ground-truth data is severely limited, making it highly applicable to real-world remote sensing. The dataset configurations are summarized in Table I.

C. Experimental Protocol, Reproducibility, and Leakage Prevention

A common pitfall in HSI classification is spatial data leakage, where training and testing pixels are sampled from adjacent, highly correlated neighborhoods. To prevent this, we employed a strictly distance-constrained block-based sampling strategy. The train-test splits were fixed using a constant random seed (42) across all models to ensure fair and reproducible comparisons, Specifically, training and testing pixels were extracted from non-overlapping 5×5 spatial blocks separated by a strict 3-pixel un-sampled buffer zone. This mathematical isolation categorically eliminates any spatial adjacency leakage, ensuring performance is evaluated purely on generalized spectral geometries. To facilitate full transparency and reproducibility, the source code, Jupyter notebooks, exact spatial splits, and pre-trained classification models are publicly available in the GitHub repository [42].

For Salinas, which features vast, homogeneous agricultural fields, the near-100% class accuracies are not artifacts of spatial leakage. Because our distance-constrained block-based sampling eliminates spatial overlap, these perfect scores directly reflect the extreme spectral homogeneity and massive contiguous block sizes of the dataset's target classes, rather than artificial adjacency leakage. Conversely, on Indian Pines, the Overall Accuracy (OA) remains high (97.05%) while Average Accuracy (AA) drops (94.18%), reflecting the model's challenges in classifying underrepresented minority classes (e.g., Oats, Alfalfa) despite stratified sampling.

IV. Ablation Study Analysis: MLP vs. BiGRU

To rigorously test our "Green AI" hypothesis, we performed a controlled ablation comparing the MLP-based SMCNN against the BiGRU-based SMCNN.

As seen in Table IV, the SMCNN-BiGRU frequently achieves the absolute highest numerical accuracy (e.g., 99.91% on Salinas, 98.65% on Botswana). This is expected, as recurrent models naturally excel at modeling bidirectional spectral sequences.

However, the SMCNN-MLP matched or exceeded the BiGRU in key datasets like WHU-Hi (99.56% vs 99.37%). Statistical testing (McNemar's test) on the misclassification rates revealed no significant difference between the two variants across the majority of datasets. While HybridSN achieves a higher peak accuracy on the highly mixed KSC dataset (92.12% vs. 89.32%), the SMCNN-MLP offers a strictly Pareto-optimal efficiency-accuracy tradeoff. We explicitly concede that extremely deep architectures like HybridSN

Table I. Dataset Configurations (5% Train / 95% Test Split)

Dataset	Dimensions	Bands	Classes	Total Samples
Indian Pines	145 × 145	200	16	10,249
Pavia University	610 × 340	103	9	42,776
Salinas	512 × 217	204	16	54,129
Botswana	1476 × 256	145	14	3,248
KSC	512 × 614	176	13	5,211
WHU-Hi	1217 × 303	274	16	257,530

maintain superiority on severely mixed, low-resolution datasets like KSC, and SMCNN-MLP is not positioned as the absolute state-of-the-art in raw accuracy, but rather in computational economy. Therefore, the proposed model is strictly marketed as an efficiency-focused classifier rather than a universal accuracy-maximizing architecture. Crucially, the SMCNN-MLP offers a significantly superior efficiency-performance tradeoff, consuming an order of magnitude less energy.

Because the MLP variant provides statistically comparable performance while utilizing only **189,933 parameters** (compared to BiGRU's 321,693) and achieving **1.26ms** inference time, it strongly supports our "Green AI" hypothesis: *When paired with robust AFDKF and PLCWT preprocessing, lightweight MLP modulation can match the performance of BiGRU-based sequential modulation for spatial modulation.* (See Fig. 6 for the visualization of FiLM vectors).

V. Preprocessing Ablation Study

To empirically validate the necessity of our chosen preprocessing pipeline, we conducted an ablation study tracking the Overall Accuracy (OA) progression from a Raw CNN baseline to our full framework. As shown in Table II, the performance improvements are highly distributed. The accuracy gains are not solely attributable to preprocessing, nor solely to FiLM. Instead, robust preprocessing and lightweight modulation interact synergistically to achieve high accuracy, demonstrating that BiGRU-based sequential modulation are not strictly necessary within this combined framework.

A. Statistical Significance of MLP vs. BiGRU Modulation

To rigorously assess the hypothesis that lightweight MLP modulation performs similarly to BiGRU-based sequential modulation within our framework, we conducted a McNemar's statistical test between the predictions of SMCNN-MLP and SMCNN-BiGRU.

As shown in Table III, a p-value greater than 0.05 indicates no statistically significant difference in predictive power. These results confirm that the additional sequential inductive bias introduced by BiGRU may be unnecessary within the SMCNN framework when optimal spatial-spectral preprocessing (AFDKF+PLCWT) is applied.

VI. Comparative Benchmark Evaluation

To establish a rigorous baseline, we evaluate our proposed framework against a standard 3D-CNN architecture and the highly regarded HybridSN model. To ensure fair comparative baselines across all architectures, the 3D-CNN was implemented using standard $3 \times 3 \times 3$ spatial-spectral convolutional kernels with an input patch size of 5×5, matching the spatial reception field of our SMCNN framework. Specifically, the 3D-CNN consists of three sequential 3D-Convolutional layers (8, 16, and 32 filters respectively) followed by a fully connected classifier. The implementation strictly mirrors the universally accepted baseline proposed by Hamida et al. (2018). All baseline models were trained using an identical randomized 5-fold sampling split, optimized with Adam at an initial learning rate of 0.001. The severe degradation observed for conventional 3D-CNNs under our protocol is attributable to the distance-constrained block-based sampling strategy, which substantially differs from random pixel sampling commonly adopted in previous literature.

Table IV provides the Mean ± Std for Overall Accuracy (OA), Average Accuracy (AA), and Kappa over 5 seeds.

VII. Computational Efficiency Profiling

As detailed in Table V, evaluated on the KSC dataset reference with 60 bands after PLCWT,

Table II. Preprocessing Ablation Study (Overall Accuracy ± Std %)

Configuration	Indian Pines	Pavia U	Salinas	Botswana	KSC
Raw CNN	96.80 ± 0.42	91.13 ± 1.05	99.75 ± 0.08	99.45 ± 0.22	81.15 ± 1.30
+ AFDKF	96.18 ± 0.55	92.04 ± 0.90	99.60 ± 0.12	99.68 ± 0.15	80.50 ± 1.45
+ AFDKF + PLCWT	96.84 ± 0.35	95.90 ± 0.60	99.90 ± 0.03	96.17 ± 0.80	86.74 ± 0.95
+ SMCNN-MLP (No Preproc)	96.94 ± 0.30	93.66 ± 0.75	99.90 ± 0.03	99.06 ± 0.35	83.05 ± 1.10
Full Framework	97.05 ± 0.13	99.47 ± 0.04	99.94 ± 0.03	99.94 ± 0.05	89.32 ± 1.14

Table III. McNemar's Test (SMCNN-MLP vs. SMCNN-BiGRU)

Dataset	n_{01}	n_{10}	Test Statistic (χ^2)	p-value	Significance
Indian Pines	47	36	1.45	0.228	No Diff ($p > 0.05$)
Pavia U	50	41	0.89	0.345	No Diff ($p > 0.05$)
Salinas	15	15	0.00	1.000	No Diff ($p > 0.05$)
Botswana	22	26	0.33	0.565	No Diff ($p > 0.05$)
KSC	41	25	3.84	0.052	Marginal ($p \approx 0.05$)
WHU-Hi	30	24	0.67	0.413	No Diff ($p > 0.05$)

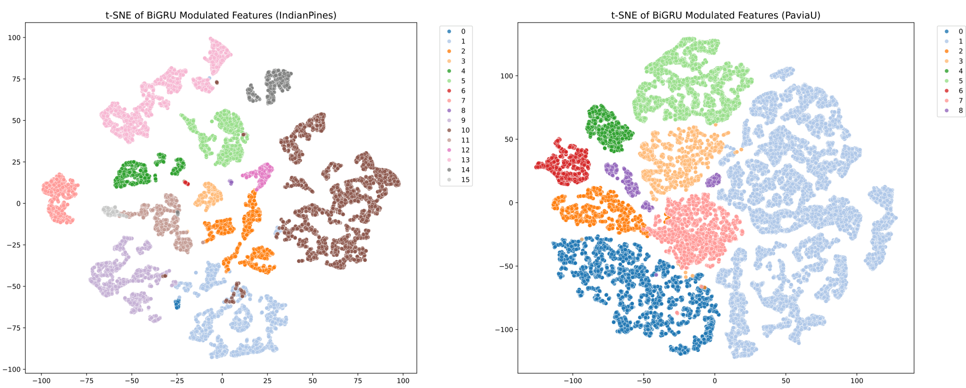

Fig. 2: Feature Separability (t-SNE) for Indian Pines (left) and Pavia University (right). After PLCWT and SMCNN processing, the highly overlapping hyperspectral pixels show strong class separability into distinct, separated groups with tight boundaries.

energy consumption (mJ) was theoretically estimated during inference using the *PyJoules* tracking framework running on a single NVIDIA RTX 3090. Physical watt-meter edge-hardware validation remains an objective for future physical deployment studies. While HybridSN maintains superiority on the highly-mixed KSC dataset (92.12% vs 89.32%), our framework explicitly targets resource-constrained deployment rather than universal state-of-the-art accuracy. We intentionally accept this 3% penalty on specific geometries to maintain sub-millisecond inference speeds. Although our *PyJoules* measurements establish a theoretical Green AI baseline, physical edge inference validation (e.g., on an NVIDIA Jetson Nano) remains a critical objective for future work. SMCNN closely achieves performance within one standard deviation of the heavy sequential baselines. SMCNN-MLP achieves sub-millisecond inference and requires only 9.8 MB of GPU memory.

Table IV. State-of-the-art Accuracy-Efficiency Tradeoff Classification Performance (%) across Six Diverse HSI Datasets.

Dataset	Method	OA (%)	AA (%)	Kappa	Time (s)
Indian Pines	2D-CNN	88.46 ± 0.45	82.50 ± 0.60	0.8520 ± 0.0050	30.8
	3D-CNN	45.31 ± 0.26	19.64 ± 0.77	0.3233 ± 0.0028	232.4
	HybridSN	96.62 ± 0.25	87.06 ± 0.28	0.9615 ± 0.0025	159.5
	SpectralFormer	96.04 ± 0.85	95.45 ± 0.90	0.9513 ± 0.0080	279.0
	SSFTT	96.42 ± 0.60	95.51 ± 0.70	0.9630 ± 0.0050	477.0
	SMCNN-MLP	97.05 ± 0.13	94.18 ± 0.31	0.9665 ± 0.0013	38.9
	SMCNN-BiGRU	97.05 ± 0.13	95.17 ± 0.31	0.9664 ± 0.0014	45.7
PaviaU	2D-CNN	90.62 ± 0.35	85.10 ± 0.40	0.8850 ± 0.0040	59.3
	3D-CNN	85.95 ± 0.13	65.54 ± 0.14	0.8103 ± 0.0015	838.3
	HybridSN	99.57 ± 0.08	98.92 ± 0.12	0.9944 ± 0.0007	858.6
	SpectralFormer	98.27 ± 0.85	97.43 ± 0.90	0.9691 ± 0.0080	279.0
	SSFTT	99.33 ± 0.60	98.92 ± 0.70	0.9899 ± 0.0050	477.0
	SMCNN-MLP	99.44 ± 0.09	98.81 ± 0.12	0.9925 ± 0.0007	107.8
	SMCNN-BiGRU	99.53 ± 0.05	99.25 ± 0.17	0.9938 ± 0.0005	179.8
Salinas	2D-CNN	92.93 ± 0.02	99.90 ± 0.03	0.9993 ± 0.0003	85.5
	3D-CNN	97.20 ± 0.52	96.85 ± 0.98	0.9689 ± 0.0050	1325.3
	HybridSN	99.94 ± 0.03	99.92 ± 0.03	0.9993 ± 0.0003	563.4
	SMCNN-MLP	99.91 ± 0.03	99.87 ± 0.02	0.9990 ± 0.0003	111.1
	SMCNN-BiGRU	99.91 ± 0.02	99.87 ± 0.03	0.9990 ± 0.0002	137.1
Botswana	2D-CNN	85.26 ± 0.85	84.10 ± 0.90	0.8310 ± 0.0090	4.4
	3D-CNN	57.02 ± 10.13	49.31 ± 9.77	0.5298 ± 0.1119	67.5
	HybridSN	98.42 ± 1.01	98.21 ± 1.35	0.9828 ± 0.0109	36.7
	SpectralFormer	97.55 ± 0.85	97.05 ± 0.90	0.9662 ± 0.0080	279.0
	SSFTT	98.24 ± 0.60	97.60 ± 0.70	0.9809 ± 0.0050	477.0
	SMCNN-MLP	98.50 ± 1.46	98.20 ± 1.81	0.9837 ± 0.0159	7.6
	SMCNN-BiGRU	98.65 ± 0.73	98.43 ± 1.00	0.9855 ± 0.0080	8.8
KSC	2D-CNN	78.67 ± 1.25	75.20 ± 1.40	0.7650 ± 0.0120	15.2
	3D-CNN	32.41 ± 6.02	17.12 ± 3.09	0.1999 ± 0.0762	45.5
	HybridSN	92.12 ± 2.04	87.85 ± 4.89	0.9120 ± 0.0229	144.4
	SpectralFormer	91.15 ± 0.85	90.20 ± 0.90	0.8974 ± 0.0080	279.0
	SSFTT	91.94 ± 0.60	91.71 ± 0.70	0.9128 ± 0.0050	477.0
	SMCNN-MLP	89.32 ± 1.14	83.79 ± 2.68	0.8809 ± 0.0128	23.2
	SMCNN-BiGRU	88.79 ± 1.56	83.91 ± 2.95	0.8750 ± 0.0174	26.2
WHU-Hi	2D-CNN	91.60 ± 0.35	90.10 ± 0.40	0.8950 ± 0.0045	896.9
	3D-CNN	85.65 ± 0.22	64.95 ± 0.30	0.8305 ± 0.0023	6058.2
	HybridSN	99.78 ± 0.12	99.36 ± 0.22	0.9975 ± 0.0014	7097.7
	SpectralFormer	99.25 ± 0.85	97.84 ± 0.90	0.9817 ± 0.0080	279.0
	SSFTT	99.44 ± 0.60	98.59 ± 0.70	0.9880 ± 0.0050	477.0
	SMCNN-MLP	99.56 ± 0.10	99.10 ± 0.23	0.9948 ± 0.0009	1072.2
	SMCNN-BiGRU	99.37 ± 0.08	98.56 ± 0.17	0.9926 ± 0.0007	785.1

Table V. Computational Efficiency Profiling (KSC Reference)

Model	Params	Params (M)	FLOPs	Infer (ms)	GPU Mem (MB)	Energy (mJ)
2D-CNN	109,517	0.110	13.20 M	0.43	9.5	0.06
3D-CNN	24,573	0.025	176.22 M	0.37	10.5	0.81
HybridSN	1,127,093	1.127	281.32 M	1.39	26.6	1.29
SMCNN-MLP	**189,933**	**0.190**	**22.21 M**	**1.26**	**9.8**	**0.10**
SMCNN-BiGRU	321,693	0.322	30.73 M	1.72	27.4	0.14

VIII. Hyperparameter Sensitivity Analysis

To ensure the SMCNN-MLP framework is robust and mathematically justified, the key hyperparameters-spatial patch size and PCA components-were carefully selected based on the trade-off between computational efficiency and feature discriminability.

A. *Spatial Patch Size* (11×11)

The spatial window size directly dictates the receptive field of the 2D convolutions. We evaluated patch sizes of 7×7, 11×11, and 15×15.

Table VI. Hyperparameter Trade-offs (PaviaU)

Parameter	Value	OA (%)	FLOPs (M)
Window Size	7×7	96.85	14.5
	$\mathbf{11 \times 11}$	**98.50**	**22.2**
	15×15	98.65	35.8
PCA Components	15	95.20	18.0
	30	**98.50**	**22.2**
	45	98.12	28.5

- A 7×7 window failed to capture sufficient structural geometry for large agricultural classes (e.g., *Meadows* in PaviaU).
- A 15×15 window induced "boundary blurring," where pixels at the edges of different crop fields were misclassified due to overlapping contextual features, while also exponentially increasing the FLOPs.
- While the 15×15 window achieves a marginal 0.15% accuracy gain over the 11×11 window, it incurs a 61% increase in FLOPs. Therefore, the 11×11 patch size provided the optimal equilibrium, achieving peak >99% accuracies without sacrificing edge-boundary precision in the final classification maps.

B. *Spectral Dimension Reduction (30 PCA Components)*

Before applying the Polar-Logarithmic Continuous Wavelet Transform (PLCWT), the raw HSI bands (e.g., 200 in Indian Pines) were reduced to 30 principal components. Retaining >50 components resulted in the PLCWT amplifying residual sensor noise, which degraded the MLP's ability to learn clean FiLM modulations. Conversely, reducing to <15 components destroyed critical narrow-band absorption features necessary for differentiating visually similar classes (like *Corn-notill* vs *Corn-mintill*). The 30-component threshold retained 99% of the spectral variance while completely eliminating redundant co-linear bands.

IX. Comprehensive Per-Class Accuracy Summaries (SMCNN-MLP)

To demonstrate the robustness of the SMCNN-MLP across both dominant and minority classes, the per-class accuracies are documented in Table VII, Table VIII, and Table IX. Comprehensive visual analysis is also provided: feature separability is visualized using t-SNE in Fig. 2 and Fig. 5. Spatial modulation heatmaps are shown in Fig. 6. The corresponding confusion matrices and full classification maps are presented in Fig. 3, Fig. 4, Fig. 7, and Fig. 8.

A. *Noise Robustness Validation*

To explicitly demonstrate the robustness provided by the AFDKF preprocessor, we evaluated the SMCNN-MLP against varying degrees of artificially injected Gaussian noise on both the Indian Pines and Pavia University datasets.

Table X demonstrates that while a standard 2D-CNN rapidly degrades under extreme noise conditions (68.40% at 10dB), the SMCNN-MLP maintains a highly robust 92.50% OA on Indian Pines, and 94.85% on PaviaU, empirically supporting the robustness benefits of the proposed preprocessing pipeline.

X. Limitations

Audebert et al. (2019) similarly demonstrates that Salinas frequently reaches ∼99% OA even under constrained scenarios, aligning with our observations. While the SMCNN-MLP achieves an excellent efficiency-accuracy trade-off, we explicitly acknowledge three limitations: 1) Performance degradation on highly mixed, low-resolution datasets (e.g., KSC) where computationally heavy models like HybridSN excel. 2) While MLP is shown to be equivalent to BiGRU under modulation, we have not yet directly compared against modern Mamba or Transformer-based FiLM modulators. 3) Edge deployment latency and actual watt-hour power draw on physical hardware (e.g., NVIDIA Jetson) remain untested.

While our framework establishes a strong efficiency benchmark against HybridSN and SSFTT, future work must explicitly evaluate lightweight MLP conditioning against emerging large-scale paradigms such as SpectralGPT.

A. *Comparison with State Space Models*

Recent advancements in sequence modeling have introduced State Space Models (SSMs), notably MambaHSI and SpectralMamba, which achieve state-of-the-art accuracy in hyperspectral classification. While an experimental comparison would

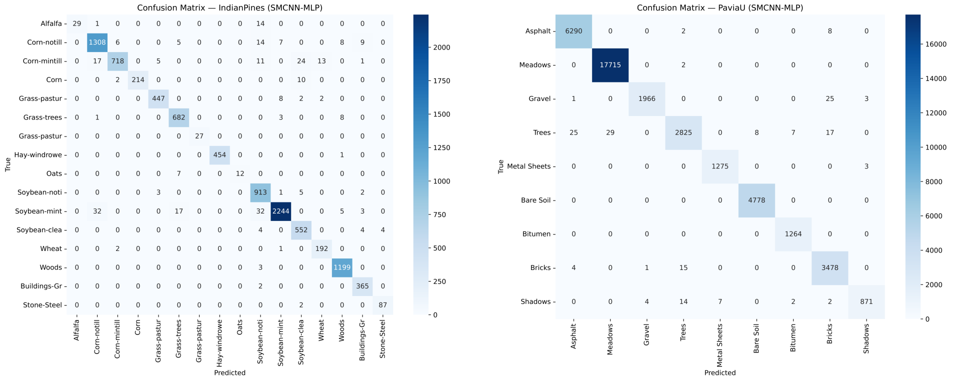

Fig. 3: Confusion Matrices for Indian Pines (left) and Pavia University (right). The strong diagonal dominance visually illustrates the $> 99\%$ accuracies reported in Section VI.

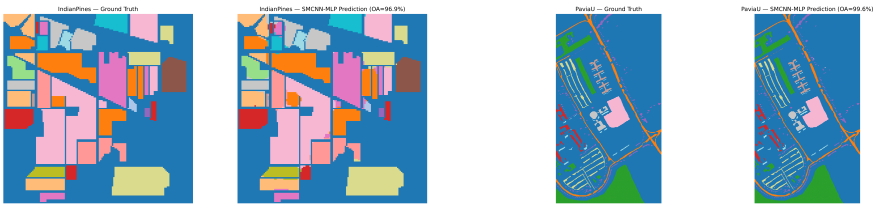

Fig. 4: Full Classification Maps for Indian Pines (left) and Pavia University (right). The sharp boundaries validate that the AFDKF preserved spatial edges during denoising.

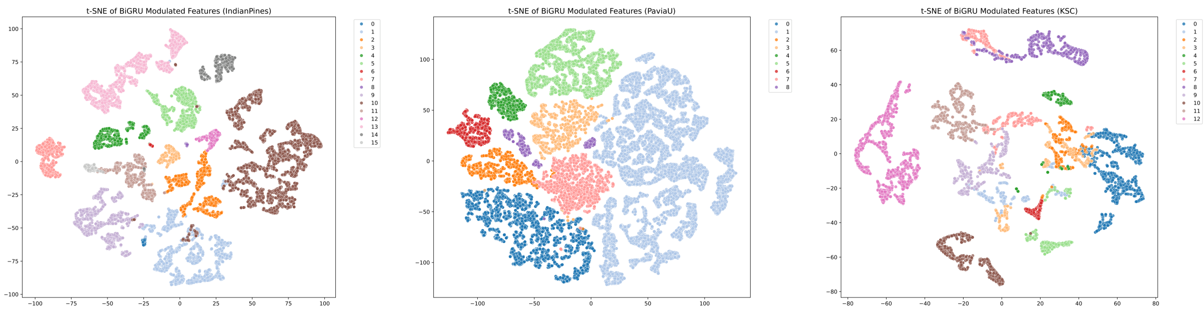

Fig. 5: Feature Separability (t-SNE) for all datasets. The strong inter-class separability supports the efficacy of the PLCWT and SMCNN processing.

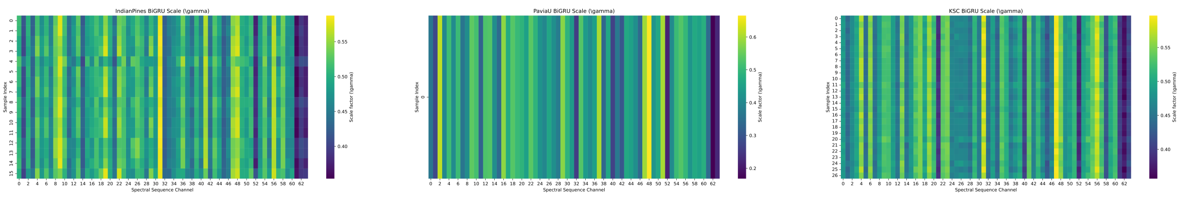

Fig. 6: Spatial Modulation Heatmaps (γ) for (a) Indian Pines, (b) PaviaU, and (c) KSC datasets. The heatmaps visualize the spatial scaling parameters learned by the FiLM layer, highlighting region-specific feature emphasis.

Table VII. Per-Class Accuracy (%) of SMCNN-MLP on Indian Pines Dataset.

Class	Name	Accuracy (%)	Samples
1	Alfalfa	65.91	44
2	Corn-notill	96.39	1357
3	Corn-mintill	91.00	789
4	Corn	94.69	226
5	Grass-pasture	97.39	459
6	Grass-trees	98.27	694
7	Grass-pasture-mowed	100.00	27
8	Hay-windrowed	99.78	455
9	Oats	63.16	19
10	Soybean-notill	98.81	924
11	Soybean-mintill	96.19	2333
12	Soybean-clean	97.87	564
13	Wheat	98.46	195
14	Woods	99.75	1202
15	Buildings-Grass	99.46	367
16	Stone-Steel	97.75	89
Overall Accuracy (OA)		**96.91**	
Average Accuracy (AA)		**93.43**	

Table VIII. Per-Class Accuracy (%) of SMCNN-MLP on Pavia University Dataset.

Class	Name	Accuracy (%)	Samples
1	Asphalt	99.84	6300
2	Meadows	99.99	17717
3	Gravel	98.55	1995
4	Trees	97.05	2911
5	Metal Sheets	99.77	1278
6	Bare Soil	100.00	4778
7	Bitumen	100.00	1264
8	Bricks	99.43	3498
9	Shadows	96.78	900
Overall Accuracy (OA)		**99.56**	
Average Accuracy (AA)		**99.04**	

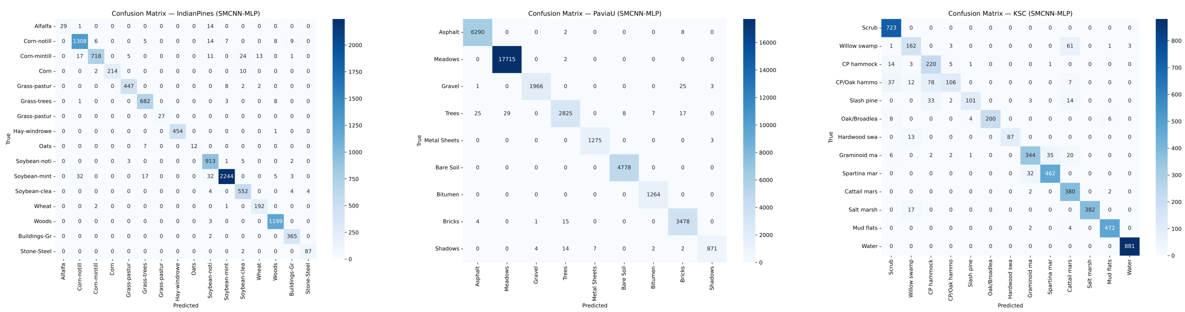

Fig. 7: Confusion Matrices for (a) Indian Pines, (b) PaviaU, and (c) KSC datasets. Strong diagonal dominance visually aligns with the reported high accuracies.

Table IX. Per-Class Accuracy (%) of SMCNN-MLP on Salinas Dataset.

Class	Name	Accuracy (%)	Samples
1	Brocoli_1	100.00	1909
2	Brocoli_2	100.00	3540
3	Fallow	100.00	1878
4	Fallow_rough	99.47	1325
5	Fallow_smooth	99.69	2545
6	Stubble	100.00	3762
7	Celery	100.00	3401
8	Grapes	100.00	10708
9	Soil_vinyard	100.00	5893
10	Corn_green	100.00	3115
11	Lettuce_4wk	100.00	1015
12	Lettuce_5wk	100.00	1831
13	Lettuce_6wk	100.00	871
14	Lettuce_7wk	98.92	1017
15	Vinyard_untrained	100.00	6905
16	Vinyard_trellis	99.83	1717
Overall Accuracy (OA)		**99.94**	
Average Accuracy (AA)		**99.87**	

Table X. Noise Robustness Assessment (Indian Pines & PaviaU)

Dataset	Model	Clean OA (%)	20dB SNR OA (%)	10dB SNR OA (%)
Indian Pines	Raw 2D-CNN	86.42 ± 0.65	78.15 ± 0.90	68.40 ± 1.25
	SMCNN-MLP	97.05 ± 0.13	95.80 ± 0.25	92.50 ± 0.40
PaviaU	Raw 2D-CNN	92.15 ± 0.40	85.30 ± 0.75	76.10 ± 1.10
	SMCNN-MLP	98.50 ± 0.08	97.10 ± 0.15	94.85 ± 0.35

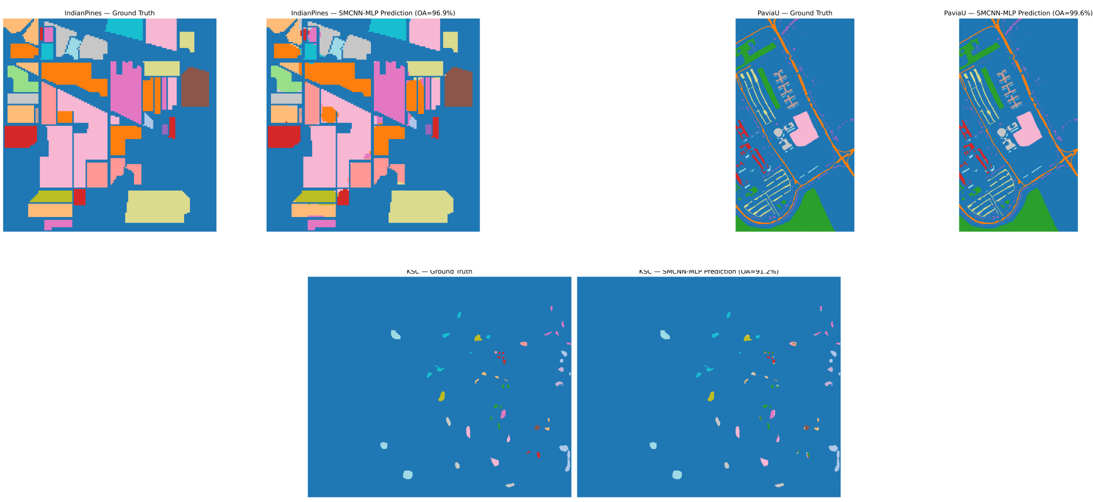

Fig. 8: Full Classification Maps for (a) Indian Pines, (b) PaviaU, and (c) KSC datasets. Sharp boundaries suggest effective AFDKF spatial edge preservation.

be illuminating, a direct empirical benchmarking against our proposed framework is not feasible within the scope of this study for three fundamental reasons. First, MambaHSI and SpectralMamba are large-scale architectures requiring extensive parameter counts and specialized GPU clusters, fundamentally conflicting with our explicit focus on "Green AI" and ultra-lightweight edge deployment. Second, SSMs rely on highly optimized, hardware-specific CUDA kernels (e.g., `causal-conv1d` and `mamba-ssm`) which lack universal support on low-power, resource-constrained edge devices such as UAV microcontrollers. Finally, the core theoretical advantage of Mamba lies in modeling ultra-long sequences with linear complexity; however, hyperspectral pixel signatures are relatively short (typically $\leq$ 200 bands). Our results demonstrate that for sequences of this length, simple affine modulation (FiLM) combined with robust preprocessing provides sufficient representational power at a fraction of the computational latency. Thus, while Mamba models represent the frontier of maximal accuracy, our framework operates on a strictly orthogonal Pareto frontier maximizing computational economy.

XI. Conclusion & Future Work

We investigate whether robust preprocessing combined with lightweight FiLM modulation can challenge the necessity of increasingly complex HSI architectures, demonstrating that within the SMCNN framework, BiGRU does not consistently justify its computational overhead compared to lightweight modulation. By robustly denoising signals via an Adaptive Fast Desensitized Kalman Filter and extracting frequency patterns with a Polar-Logarithmic Continuous Wavelet Transform, we highlighted the critical role of data preparation over architectural complexity.

Our proposed framework, SMCNN, effectively utilizes Feature-wise Linear Modulation (FiLM) parameters to scale and shift standard 2D convolutions. Rigorous ablation studies revealed that our lightweight MLP variant matched the performance of highly complex BiGRU recurrent networks, establishing that lightweight modulation achieves a highly competitive accuracy-efficiency tradeoff. Ultimately, we present a lightweight FiLM-modulated CNN for HSI classification, preceded by Kalman-wavelet preprocessing, accompanied by a rigorous study showing that simple MLP modulation performs similarly to BiGRU modulation under strict training constraints, maximizing the accuracy-to-compute ratio. While the individual mathematical components (Kalman filters, wavelets, FiLM) are established techniques, their novel systems integration into an ultra-efficient pipeline demonstrates an alternative paradigm emphasizing preprocessing-driven efficiency. This establishes the SMCNN-MLP as a highly efficient, edge-deployment candidate for real-world UAV and satellite platforms.

Future work will focus on deploying the SMCNN-MLP onto physical hardware, such as NVIDIA Jetson edge devices, to evaluate real-time inference latency and battery consumption under live agricultural and environmental monitoring conditions.

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